

Max Conway

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PROFESSIONAL EXPERIENCE

University of Colorado Boulder

Boulder, CO

Teaching Assistant

Aug 2024 – Dec 2024

Research Assistant

Aug 2024 – Now

- Led a 17-person team developing GOLEM, a modular humanoid autonomy stack enabling high-precision industrial tasks including EV battery disassembly at millimeter-level accuracy (golem-humanoid.github.io)
- Built digital twin environments in Isaac Lab and MuJoCo for sim-to-real policy training, reducing hardware iteration cycles during development
- Designed and deployed a robotic arm platform with linear actuator for autonomous EV battery disassembly, owning motion planning, perception, and mechanical design; published as second author (CASE 2026)

University of Denver

Denver, CO

Adjunct Professor

June 2024 – Aug 2024

Research Assistant

Sept 2022 – Aug 2024

Teaching Assistant

Nov 2023 – June 2024

- Developed and taught a course on applied machine learning that taught students machine learning theory, including regressions, gradient descent, back propagation, multilayer perceptrons, and transformers. By the end of the class students had developed image classification and reinforcement learning projects in PyTorch.
- Teaching assistant for four upper level computer science classes: Machine Learning, Software for AI Robotics, Embedded Systems, and Advanced Algorithms. All of these classes I previously excelled in and loved.
- Researched multimodal sensor fusion across thermal, depth and RGB cameras for human activity recognition at the ARISE robotics lab (arise-lab.com). Supervised by Dr. Christopher Reardon. Resulted in a paper that was accepted to IROS 2024

DEVCOM Army Research Laboratory

Adelphi, MD

Masters Research Intern

June 2023 – Sept 2023

- Worked on fundamental research problems in the intersection of computer vision and knowledge representations. My work at ARL was focused on perception for autonomous robotics. This work continued from a summer internship into a collaboration between ARL and the University of Denver.

Amazon, Inc.

Remote from Denver, CO

Software Engineer Intern

June 2022 – Sept 2022

- Designed, coded and delivered improved notification experience for multimedia Alexa devices. Wrote design document, learned new technology stack and worked with UX team to complete internship project ahead of schedule. Received offer to return as full time software engineer.

EDUCATION

University of Colorado Boulder,

Boulder, CO

Candidate for PhD in Computer Science

Expected to finish in 2028

GPA: 4.0

- **Academics:** Dean's & Departmental Excellence Fellowship
- **Relevant Coursework:** Advanced Computer Vision, Transformers in Robotics, Decision Making under Uncertainty, Reinforcement Learning, Vision Language Models for Robotics

University of Denver, Ritchie School of Engineering and Computer Science

Denver, CO

M.S. & B.S. in Computer Science. Minors: Physics & Math

Masters complete December 2023

M.S. GPA: 4.0, B.S. GPA: 3.85

- **Academics:** Dean's List, Hornbeck Scholar.
- **Relevant Coursework:** ROS, Machine Learning, Software for AI Robotics, Programming Languages (including C++, Java, Python), Compilers, Computer Organization

PUBLICATIONS

- Max Conway, Brian Reily, Christopher Reardon: **Learned Sensor Fusion For Robust Human Activity Recognition in Challenging Environments** (Accepted to IROS 2024)
- Andrew Hannum, Max Conway, Andre Harrison, Mario Lopez: **Data-Driven Hierarchical Open Set Recognition** (Accepted to ICRA@40)
- Zachary Allen, Max Conway, Lyle Antieau, Allen Ponraj, Nikolaus Correll: **Robotic Agentic Platform for Intelligent Electric Vehicle Disassembly** (Accepted to CASE 2026)
- Max Conway, et al: **GOLEM: Modular Humanoid Autonomy Towards Electric Vehicle Battery Disassembly** (Under Review at IEEE-RAS Humanoids 2026)